

Andrew Du

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TECHNICAL SKILLS

Hardware: Altium, Vivado, LTSpice, PCB Design, FPGAs, MCUs (STM32, RPi, Arduino), I2C, SPI, UART, CAN, JTAG

Software: Verilog, Python, C/C++, Java, MATLAB, Linux, Git

Lab Tools: Oscilloscopes, Vector Network Analyzer, Function Generator, Spectrum Analyzer, Logic Analyzer, Multimeter

PROFESSIONAL EXPERIENCE

Tesla - Vehicle Hardware Intern

May 2025 - August 2026

- Designed, laid out, and brought up **eight 8-to-12-layer Processor-in-Loop (PIL) PCBAs** used to emulate multiple in-vehicle ECUs for Tesla's software validation stack.
 - Planned pin assignment on **400+ pin BGA processors** and performed routing for hundreds of I/O including impedance-matched high-speed RGMII, LVDS SerDes, USB, CAN, and sensitive analog signals.
 - Implemented high-speed multiplexing of signals into shared processor pins and integrated PIL boards with a **K26 SoM** (FPGA + ARM core) to facilitate emulation of diverse test cases and multiple ECU variants.
 - Led board bring-up: used oscilloscope and vector network analyzer to verify power rails and characterize signal integrity (propagation delay, rise-time, bandwidth) across multiple high-speed switches and analog ICs.
- Designed and testbenched RTL behavioral models of five TI motor-driver and LED-driver ICs, implementing register maps with reset defaults, register read/write access, and command decode LUTs for ECU emulation.
- Designed and testbenched a UART controller in RTL to handle communication between validation software and emulated peripherals.

UBC AeroDesign - Avionics Hardware

September 2023 - Present

- Designed and manufactured avionics hardware:
 - A wiring hub PCB to consolidate power distribution, sensor modules, and peripheral systems.
 - A power distribution buck converter PCB using a LM5146 controller chip; responsible for power efficiency, component selection, schematic, layout. Employed oscilloscopes and digital multimeters for performance characterization.
- Wrote C firmware for **STM32** and RTK GNSS module for communication with plane.

PROJECTS

PCB Business Card

- Designed a custom PCB that serves as an **NFC-enabled** business card using a Class 4 spiral antenna.
- Characterized antenna properties such as resonant frequency, inductance, and Q factor using a vector network analyzer.

Autonomous Racing Robot

- Designed and hand-soldered h-bridge circuits for **motor control**, infrared sensing circuits, collision detection circuits, and IMU.
- Implemented **PID control** algorithm for steering and speed optimization using STM32 and created a state machine based on sensor data inputs from sections of racetrack.

Closed-Loop Motor Speed Controller

- Designed a feedback control circuit to adjust the speed of a motor using digital logic counters, clock pulse generators, and digital-to-analog converters.
- Used logic analyzers, oscilloscope for bring-up and debugging.

EDUCATION

Engineering Physics, University of British Columbia

Graduation: April 2027

- Relevant Coursework: Signals and Systems, Electronic Circuit Design, Microcontrollers and Digital Systems.